The Average Bullshittings of Machines: Stylistic Fingerprints in Cold LLM Creative Output

A v1 empirical write-up from *An Assay of Bullshit*

Author: Bo Chesterton
Primary analyst: GPT-5.5 Thinking / Extended
Repository: https://github.com/quumble/An-Assay-of-Bullshit
Date: May 7, 2026

Abstract

Bullshit is usually treated as the waste product of large language models: content generated without stakes, context, or commitment. This experiment treats that waste product as an empirical signal. In a blind creative-writing assay, six models answered the same nine cold prompts across three genres and three length conditions, with ten runs per cell. The resulting 540 responses were coded with parallel semantic and structural heuristics.

The central result is that model identity is visible less in what the models write about than in how they habitually arrange what they write. Title lines, block structure, preambles, paragraphing, punctuation, and output-budget pressure form stable stylistic fingerprints. Provider family matters: GPT-family outputs were usually single-block prose, while Claude-family outputs were much more often titled and formatted. But that family story is too blunt. The sharper result is model-specific: Sonnet, Haiku, Opus, GPT 5.4, GPT 5.4 Mini, and GPT 5.4 Nano each leave a different trail in the average bullshit they produce.

A prompt-blocked classifier using only surface-style features identified the six individual models at 49.8% accuracy, compared with 16.7% chance. This is not a production classifier and not a claim about intrinsic quality. It is a sign that the models have recognizable defaults. Even when prompted to do the same arbitrary thing, they do not bullshit in the same way.

---

1. Introduction: nonsense as instrument

A cold prompt is not a natural habitat for intelligence. It is closer to a vacuum chamber. Remove conversation, task history, domain stakes, and user personality, and the model still has to do something. That something is easy to dismiss as mere filler: a short story, an animal description, a poem, a few charming sentences arranged around no real need.

But filler has form. Defaults are not nothing. When machines are asked to improvise without context, they expose the little grooves cut into them by pretraining, post-training, product taste, instruction following, and whatever invisible etiquette has been made to feel “helpful.” The outputs may be bullshit in the

Frankfurt-adjacent sense: indifferent to external truth conditions, not necessarily false, but unmoored. Yet across hundreds of such unmoored outputs, patterns appear.

The empirical question is therefore modest and strange: **can the average bullshittings of machines reveal stable model signatures?**

The answer in this dataset is yes. The strongest signatures are not deep semantic themes. They are manners: whether the model gives the thing a title, whether it writes in a single prose block, whether it prefaces the artifact with meta-commentary, whether it uses em dashes, whether it keeps elaborating until the experimental harness cuts it off. These are not grand cognitive truths. They are behavioral fingerprints.

---

2. Design and data

The public repository describes an exploratory comparison of creative-writing defaults across Claude and OpenAI model tiers under cold, contextless prompts. The design used six models, nine prompts, ten runs per model-prompt cell, temperature 1.0, `max_tokens=400`, and no system prompt, for 540 total calls.

The six de-blinded models were:

- `gpt-5.4`
- `gpt-5.4-mini`
- `gpt-5.4-nano`
- `claude-opus-4-7`
- `claude-sonnet-4-6`
- `claude-haiku-4-5-20251001`

The prompt grid crossed three categories with three length conditions:

- **category:** story, animal, poem
- **length condition:** fewer than 5 sentences, fewer than 10 sentences, open-ended

Two coded-output tables were analyzed. One contained semantic/content heuristics; the other contained structural/form heuristics. Rows were joined by `output_id`. The joined dataset contained 540 aligned outputs, with no duplicate output IDs and no row mismatches across shared source fields.

The analysis was originally blind. After blind coding, private label maps were supplied and used to de-blind the six model labels. The resulting structural-to-semantic crosswalk was exact:

Structural label	Semantic label	Model
M1	Model A	`gpt-5.4`
M2	Model B	`gpt-5.4-nano`
M3	Model C	`claude-opus-4-7`

Structural label	Semantic label	Model
M4	Model D	`claude-sonnet-4-6`
M5	Model E	`gpt-5.4-mini`
M6	Model F	`claude-haiku-4-5-20251001`

---

3. Measures

The analysis used two families of variables.

Structural features captured the visible shape of the response: word and character counts, sentence count, paragraph count, line count, single-block formatting, title-line presence, meta-preamble presence, punctuation counts, sentence-length measures, line starts, repeated openings, and related surface cues.

Semantic features captured coarser content patterns: dominant setting, mood, plot arc, imagery categories, ending valence, tense, person, pronoun counts, animal/entity detection, formulaicness, and type-token ratio.

The distinction matters. Structural variables are mechanically legible and often sharply separated the models. Semantic variables are more interpretive and may be underpowered here. A weak semantic result could mean that the models were semantically similar under these prompts, or that the heuristics were too coarse to see richer differences.

The variable `is_truncated_400ish` is described here as an **output-cap hit**, not as natural truncation. It indicates that a response reached the experiment's fixed response budget and was cut off by the harness. It is therefore a censoring variable and a measure of latent elaboration pressure, not a failure to complete.

---

4. Results

4.1 Model-level descriptive profile

The model profiles are immediately distinct.

Model	Family	Words mean	Title	Single block	Preamble	Cap hit	Compliance
GPT 5.4	GPT	109.7	2.2%	64.4%	0.0%	5.6%	100.0%
GPT 5.4 Nano	GPT	119.4	5.6%	66.7%	0.0%	11.1%	100.0%
GPT 5.4 Mini	GPT	116.7	22.2%	65.6%	0.0%	8.9%	100.0%

Model	Family	Words mean	Title	Single block	Preamble	Cap hit	Compliance
Claude Opus 4.7	Claude	108.6	75.6%	23.3%	1.1%	22.2%	96.7%
Claude Sonnet 4.6	Claude	112.2	98.9%	0.0%	37.8%	0.0%	98.3%
Claude Haiku 4.5	Claude	119.1	100.0%	0.0%	0.0%	1.1%	90.0%

The most visible contrast is the title/block axis. GPT-family models usually produced single-block prose and rarely used titles. Claude-family models, especially Sonnet and Haiku, usually produced titled, line-broken artifacts. Yet the individual models matter more than a simple family slogan permits. Opus was less rigidly formatted than Sonnet or Haiku. GPT 5.4 Mini was more title-prone than the other GPT models. Haiku's compliance weakness was local and sentence-threshold dependent, not a Claude-family universal.

Figure 1. Model-style fingerprint heatmap. The standardized feature heatmap shows that the result is not a single family gradient. It is a fingerprint panel.

4.2 The title/block axis

The clearest two-variable view is title-line rate against single-block rate.

Figure 2. The title/block axis. GPT models cluster toward high single-block and low title-line use. Sonnet and Haiku sit at the opposite pole. But the exceptions are the interesting part. Opus is not Sonnet. GPT Mini is not GPT 5.4. The family labels describe the broad geography; the individual models are the terrain.

4.3 Strongest statistical discriminators

Categorical structural variables produced very large model-level differences.

Feature	p	Cramér V
Title-line rate	4.52e-80	0.84
Single-block rate	4.92e-43	0.62
Meta-preamble rate	7.38e-36	0.57
Output-cap-hit rate	1.73e-07	0.27

Continuous variables also differed by model, especially paragraphing and punctuation.

Feature	p	ε^2
Paragraph count	6.80e-17	0.15

Feature	p	ε^2
Semicolon count	4.76e-13	0.12
Em-dash count	9.75e-12	0.10
Colon count	0.007	0.02
Type-token ratio	0.007	0.02
Formulaicness	0.068	0.01

These p-values should be read as screening statistics, not final inferential proof. The rows are not fully independent: each model answered the same prompt grid. The effect sizes and visual separations are more important than the exact p-values.

4.4 Output-cap hits: Opus had more to say

The output-cap variable needs special care. A cap-hit response did not "choose" to stop badly. It was stopped by the experiment. Therefore the strongest interpretation is not that Opus failed more often; it is that Opus more often tried to continue beyond the fixed response budget.

Figure 3. Output-cap hits by model and prompt condition. Open-ended conditions carried most of the output-cap hits.

Model	story_open	animal_open	poem_open
GPT 5.4	50.0	0.0	0.0
GPT 5.4 Nano	50.0	50.0	0.0
GPT 5.4 Mini	50.0	30.0	0.0
Claude Opus 4.7	90.0	90.0	20.0
Claude Sonnet 4.6	0.0	0.0	0.0
Claude Haiku 4.5	0.0	10.0	0.0

`claude-opus-4-7` had the highest overall cap-hit rate at 22.2%. In this design, that means Opus most often had more to say than the harness allowed. This matters analytically because cap-hit rows contaminate downstream features involving endings, plot resolution, final sentence structure, and compliance. They are informative, but they are censored observations.

4.5 Individual signatures are not reducible to family signatures

A family summary is useful but dangerous. GPT-family outputs had a 10.0% title-line rate and a 65.6% single-block rate. Claude-family outputs had a 91.5% title-line rate and a 7.8% single-block rate. That is real. It is also incomplete.

The more interesting evidence is the shape of the individual model fingerprints. In a standardized feature space built from style variables, some same-family models are farther apart than some cross-family pairs.

Figure 4. Pairwise distance among model fingerprints. The distance matrix makes the main point: family is a visible axis, but it is not the whole structure. Opus can be closer to GPT-style profiles than to Sonnet on some features, while Sonnet and Haiku share title-heavy formatting but diverge in preambles, compliance, and cap-hit behavior.

Nearest fingerprint pairs:

Model 1	Model 2	Relation	Distance
GPT 5.4	GPT 5.4 Mini	same family	2.61
GPT 5.4 Nano	GPT 5.4 Mini	same family	3.73
GPT 5.4 Mini	Claude Opus 4.7	different families	4.10
GPT 5.4 Mini	Claude Haiku 4.5	different families	4.48
GPT 5.4	GPT 5.4 Nano	same family	4.72
GPT 5.4	Claude Opus 4.7	different families	4.74

Farthest fingerprint pairs:

Model 1	Model 2	Relation	Distance
GPT 5.4 Nano	Claude Haiku 4.5	different families	5.86
GPT 5.4	Claude Sonnet 4.6	different families	6.04
Claude Opus 4.7	Claude Sonnet 4.6	same family	6.10
GPT 5.4 Nano	Claude Sonnet 4.6	different families	7.17

Figure 5. Family summaries with individual models exposed. The family means are not false. They are simply too smooth.

4.6 A prompt-blocked classifier sanity check

To test whether the stylistic fingerprints survive prompt variation, I trained simple classifiers on surface-style features only, excluding model labels, family labels, prompt labels, and the output-cap indicator. Validation was leave-one-prompt-id-out: train on eight prompt conditions, test on the ninth.

Classifier	Validation	Six-way accuracy	Chance
Logistic regression	leave-one-prompt-id-out	42.6%	16.7%
Random forest	leave-one-prompt-id-out	49.8%	16.7%

Chance performance for six-way model identification is 16.7%. The random forest reached 49.8%; logistic regression reached 42.6%. This is not offered as a serious attribution system. It is a sanity check. If a small pile of surface features can identify the model far above chance on an unseen prompt condition, then the style signal is real enough to survive the prompt grid.

4.7 Semantic features were quieter

The semantic features did not separate models as dramatically as structural variables. Formulaicness differed descriptively: `claude-opus-4-7` had the lowest mean formulaicness score, while `gpt-5.4` had the highest. But the model-level test for formulaicness was weaker than for paragraph count, title use, preambles, em dashes, or semicolons.

This should not be overread. It does not prove that the models are semantically alike. It may simply mean that the semantic coder is too blunt for the thing we want to know. Mood labels, imagery buckets, and plot-arc categories are high-level bins. A model may differ in metaphor texture, cliché handling, narrative tempo, or conceptual surprise without moving these heuristics much.

The first clean conclusion is therefore structural: **model identity is easier to see in manners than in meanings.**

4.8 Compliance and sentence-count fragility

Compliance was high overall. GPT-family models were fully compliant under the semantic heuristic on constrained prompts. Claude-family compliance was lower, but this was not a clean family effect. The weakest compliance rate was concentrated in `claude-haiku-4-5-20251001`, especially around `story_lt5`.

This is also where coder fragility matters. Sentence-count coders disagreed on 105 of 540 rows, mostly in open-ended stories and animal descriptions. Compliance coders disagreed on only three rows, all `claude-haiku-4-5-20251001` on `story_lt5`, where one sentence-count heuristic yielded compliance and the other did not. In other words, the compliance finding exists, but it is local, mechanical, and partly definition-sensitive.

---

5. Discussion: finding truths in average bullshit

The joke is also the result. Ask the machines to make something up, ask them again and again, then average the visible habits. Truth appears not in the propositions but in the posture.

The models are not merely sampling content. They are sampling manners. A title is a habit. A preamble is a habit. A refusal to put everything in one block is a habit. An em dash is a habit. So is the urge to keep elaborating until the budget cuts you off. None of these features alone is profound. Together they form a behavioral signature.

The major finding is therefore not “Claude versus GPT,” although that contrast is visible. The stronger finding is:

Individual models leave stylistic fingerprints that survive prompt variation.

Provider family is one layer of explanation. Product tier, post-training taste, instruction-following style, default helpfulness rituals, and model-specific generation preferences are probably entangled with it. This experiment cannot separate those causes. But it can show the consequence: the average output is not neutral.

The phrase “average bullshit” matters because these prompts do not ask for expertise. They ask for improvisation. The resulting text is not useless because it lacks a real-world task. It is useful precisely because the stakes are low. Low-stakes generation exposes defaults that high-stakes instruction may conceal.

---

6. Limitations

This is a v1 exploratory analysis, and its claims should stay proportional.

First, the sample is small: 10 runs per model-prompt cell. The effects are large enough to see, but replication is needed.

Second, the task regime is narrow. These are creative-writing prompts under cold contextless conditions, not coding, reasoning, retrieval, summarization, dialogue, or tool use.

Third, model names are dated product labels. The repository itself notes that dated snapshots are important for long-term reproducibility.

Fourth, rows are not fully independent because every model answered the same prompt grid. Statistical tests are best treated as descriptive screens. A paper-grade follow-up should use prompt-blocked or mixed-effects models.

Fifth, output-cap hits are censored observations. They should not be interpreted as natural failures or completion defects. They show pressure against the imposed response budget.

Sixth, structural and semantic features are heuristic. The structural coder disclosed that its script was authored by `claude-opus-4-7`, one of the experimental subjects. Many structural features are plain counts, but the authorship caveat should remain visible, especially for any finding involving Claude models.

Finally, stylistic fingerprints are not quality judgments. A title-heavy model is not better or worse than a prose-block model. A model that has more to say is not necessarily wiser. The assay measures defaults, not virtue.

---

7. Conclusion

The experiment began with disposable prompts and ended with durable signatures. The models were asked for small invented things: stories, animals, poems. They obliged. But in obliging, they revealed preferences. Some boxed the artifact with a title. Some flowed in prose. Some prefaced. Some punctuated. Some kept going until the cap fell.

The finding is not that machines bullshit. Everyone knew that. The finding is that they bullshit differently, and consistently enough to be measured.

A v1 headline:

> **Model identity is more visible in style than in semantics. Family matters, but the individual model leaves the fingerprint.**

Or, less politely:

> **There are truths in the average bullshittings of machines.**

References

- Bo Chesterton / quumble, *An-Assay-of-Bullshit*, GitHub repository: https://github.com/quumble/An-Assay-of-Bullshit
- Analysis files supplied in-session: structural coded outputs, semantic coded outputs, summary tables, and private blind-model maps.